

Adroit Technologies

Protocol Driver

Name	OPTO 22 OPTOMUX Serial
DLL	OPTO22



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1. Overview

Supports OPTOMUX slave digital and analog controllers. Each OPTOMUX board consists of a brain board with microprocessor and an I/O mounting rack. The units communicate over a RS422/485 network which interfaces to the computer via an interface converter.

This document refers to the microprocessor and I/O rack as the remote module.

2. Revisions

2.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1		K. Robinson	Changes to format, updated image resources.	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

2.2. DLL Revision

Rev.	Date	Changes
	02-Apr-1998	Initial revision
	06-Apr-1998	Fixed spelling and explanation of addressing structure
	09-Apr-1998	Changed erroneous references to module to controller.
	16-Aug-2010	Add Moxa Cluster Support

3. Wiring

PLC 25 pin		PC 25 pin
1	— Shield	1
2	—	3
3	—	2
4	]	4
5	]	5
6	]	6
8	]	8
20	]	20
7	—	7

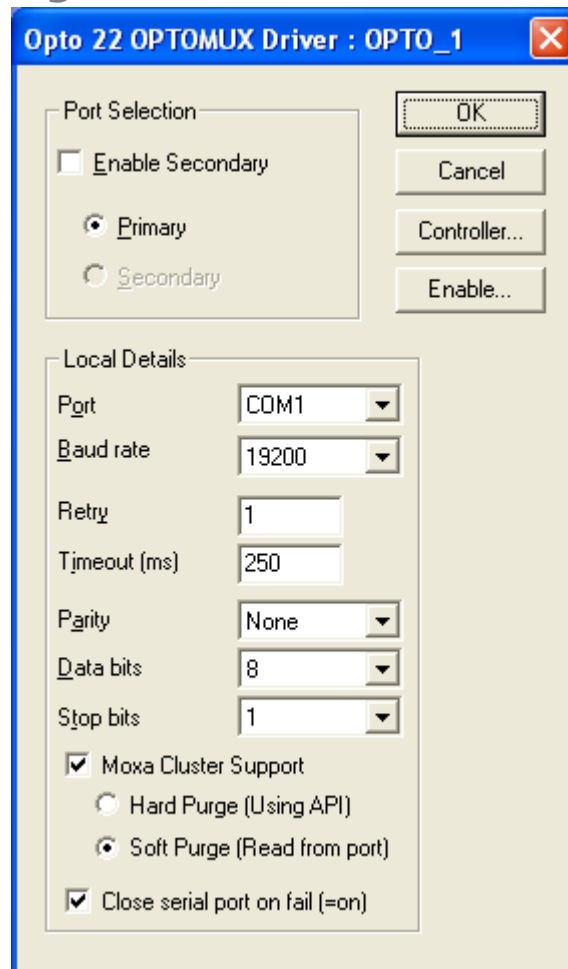


PLEASE NOTE

Shield should not be connected to ground and should only be connected at one point to prevent ground loops.

4. Device Configuration

4.1. Device Configuration



4.1.1. Port Selection

Enable Secondary – Enables secondary configuration for dual redundant Controllers. This setting is disabled for this driver.

Primary / Secondary – Select the primary or secondary button depending on which connection you wish to configure.

4.1.2. Local Details

Port – The available serial ports on your machine

Baud rate - ????

Retry – The number of times that Adroit will try to gain communication with a module before failing a transaction.

Timeout (ms) – The time that Adroit will wait for a valid response from a module before failing a transaction (Read or Write).

Parity – Default to none.

Data bits – Default to 8.

Stop bits – Default to 1.

Moxa Cluster Support - Check this option when running on a MOXA virtual port in a cluster environment. It will allow to purge the port when in standby and in shadow mode, as the MOXA unit stops functioning in this mode. Data keeps filling the buffer seeing that the driver does not do any reading from the port and eventually becomes unresponsive!

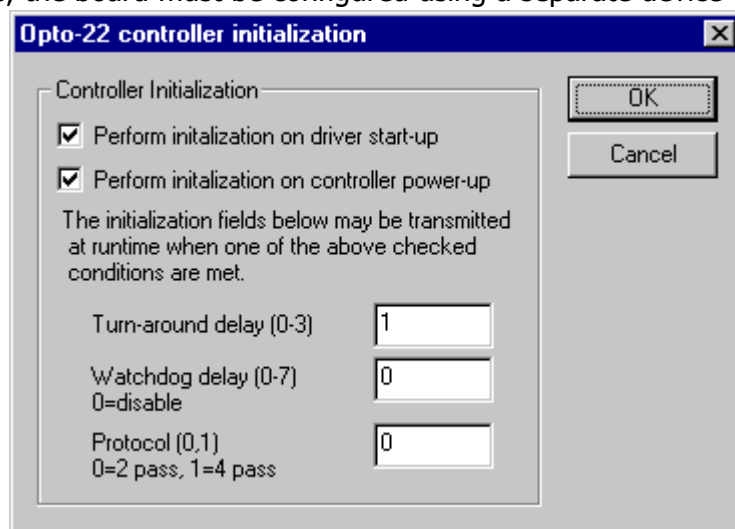
Hard Purge (Using API) - Does a call to the COM PORT to purge all the data on it. (Preferred Method).

Soft Purge (Read from Port) - Continue to read data from the port and just discard it.

Close serial port on fail - With a transaction failure closes the port and re-opens it by default but for some virtual ports it becomes an issue so we allow not closing the port in this case.

4.1.3. The Controller Configuration Dialog

The same parameters are used for all brain boards on this logical device. If different boards must use other parameters, the board must be configured using a separate device on the same com port.



Perform Initialization on driver start-up - To set whether the driver will automatically do a power-up clear when the device is started for the first time.

Perform Initialization on controller power-up – To set whether the driver will perform power-up clear when the controller restarts.

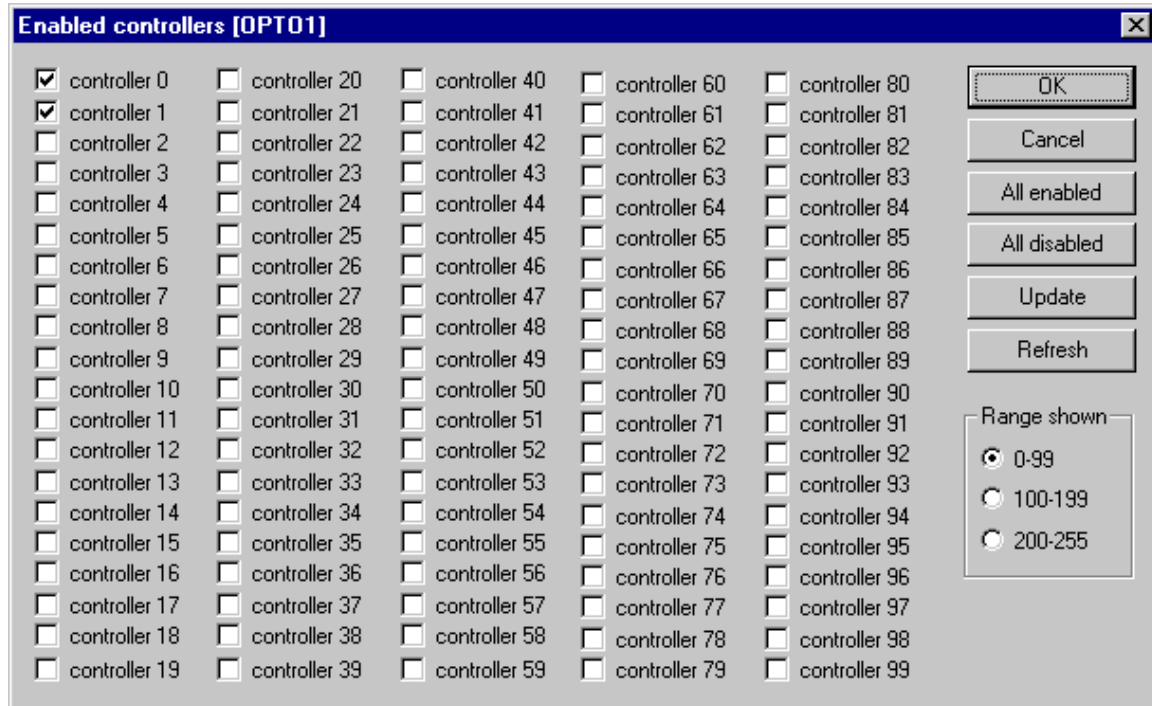
Turn-around delay- 0=no delay, 1=10 milliseconds, 2= 100 milliseconds, 3= 500 milliseconds.

Watchdog Delay 0= disable,
 1= 10 seconds then all outputs off
 2= 1 minute then all outputs off
 3=10 minutes then all outputs off
 4= disable
 5= 10 seconds then position 0 on all other outputs off
 6= 1 minute then position 0 on all other outputs off
 7=10 minutes then position 0 on all other outputs off

The watchdog delay parameters will be set for all module positions.

Protocol – Use 2-pass or 4-pass protocol, the driver currently only supports the 2-pass protocol.

4.1.4. The Enable Dialog.



Controllers 0 through 255 can be displayed by selecting the visible range from the range radio button group. The dialog may be used to enable or disable modules, alternately the same status can be achieved by scanning Digitals to the enable tags for the modules and displaying them on a mimic.

OK and **Update** will save the settings (the driver will then read them immediately if it is running)

Refresh will fetch the currently saved settings (this will show changes the driver may have made)

All enabled and **All disabled** will update the dialog, without saving the values.

5. Supported Addresses

5.1. General Addressing Structure

Remote I/O items are addressed by specifying the type of item (analog in/out, digital in/out), the address of the module (0-255) and the position or offset in that module. Valid I/O positions are 0 through 15. Addressing is in decimal.

Addresses are specified as follows:

Device data type	Type	Start	End	Boolean	Integer	Real
Refresh Controllers	1	NA	NA	UPDATE		
Controller Enable	2	0	255	ENABLE 0		
Digital Input	3	0	15	M 0 DI 0	M 0 DI 0	
Digital Output	4	0	15	M 0 DO 0	M 0 DO 0	
Analogue In	5	0	15		M 0 AI 0	M 0 AI 0
Analog Out	6	0	15		M 0 AO 0	M 0 AO 0

Modules are addresses by 'M' for module, followed by the controller address, the module type, then it's position from 0(zero) to 15.

Digital I/O can be scanned into a multistate, by specifying the first digital module in the controller, as input or output, if output is used, all positions will become outputs. When writing to the value, only the output modules on the board get updated.

Refresh and controller-enable are header items, i.e. they are not actual I/O points. They do however count as scan points.

All non-configured positions are assumed by the driver to be inputs.

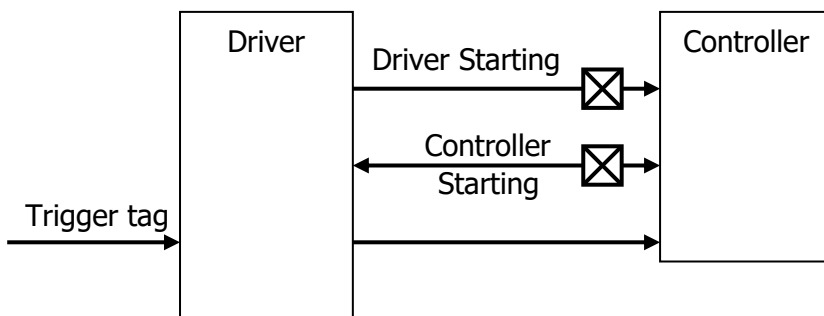
5.2. Header Items

- The special address mnemonic for a Boolean or Digital slot is supplied to initiate configuration refresh on demand: "UPDATE". Issuing a digital write with a value of 1 will initiate steps 2 & 3 of the startup sequence described in Operation. This address will return zero if scanned. This will refresh the configuration of all the enabled controllers.
- The special address mnemonic "ENABLE n" will allow enabling and disabling of scanning for a specific module "n", as an enable tag. This virtual variable can also be driven from the driver setup dialog. If the dialog is used to change the value, the digital tag (if scanned) associated with the variable will also update. A value of 1 = enabled, and zero (checkbox off) = disabled. The enable tags get saved in the system registry database for each device, if the device gets deleted in the setup program, this information is thus lost.

5.3. Operation

The driver scans all input and output variables as required by the scan agents. When writing, the driver only writes to items configured as outputs. Digitals are written using read-mask-write. On the relevant start-up flag the driver will upload all settings, a dummy force tag is also provided to allow refreshing of the configuration data. Start-up flags triggered by events are controller start-up and driver loading, providing 3 ways to refresh the controller configuration.

The controller configuration is controlled by the driver, which uses the complete I/O list of scan-points to configure the controller I/O points. If 2 scanned points for the same I/O module have conflicting information, the last one's configuration will be used. Since whichever scan point is the overriding one is not configurable, this technique is advised against. The figure below illustrates the possible ways to initialize the controller configuration.



Start-up involves 3 steps:

1. Clear power-on flag (in event of power on).
2. Set Turn-around delay, watchdog and protocol type parameters.
3. Send all I/O configuration information to the controller.

All Opto22 controller I/O configuration data will be derived from actual scanning configured in the adroit database, so as a new scan-point gets configured it will be down-loaded to the controller. Already configured scan points all get sent immediately when the scanning starts (i.e. device starts) as the system loads.

5.3.1. Multiple Devices

Multiple devices may not be assigned to scan the same controller, else the I/O configuration information which is down-loaded will conflict. If multiple devices are used to address the boards on a single line, they must be uniquely assigned to scan the boards.

	Rack 0	Rack 1	Rack 2	Rack 3	Rack 4	Rack 5	Rack 6	Etc..
Device#1	X		X	X	X			
Device#2		X				X	X	

To scan boards on a separate line, a new device must be assigned to that line, the above rule will not apply to devices on separate communication ports or lines.

5.3.2. Tag Scaling Configuration

Opto22 modules provide a raw output range of 0-4095, the actual value derived to the SCADA is biased with a constant value of 4096, thus the raw range of an analog is 4096 to 8192. Missing modules and modules not already configured in the scanning database, will return 0 (zero). All slots in a module, which are not configured, are assumed to be inputs.

6. Protocol Definition and Considerations

The protocol driver monitor Datascope can be run in ASCII mode to analyse protocol messages. All commands are followed by a carriage return, which is only shown in the HEX mode diagnostic display.

The nibbles in the position field and any bit-fields can be interpreted from right-to-left using the following table: i.e. "8000" means bit 15 on, and "0003" means bit 0 and 1 are on.

6.1. Position Map

Hex ASCII	Binary			
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
A	1	0	1	0
B	1	0	1	1
C	1	1	0	0
D	1	1	0	1
E	1	1	1	0
F	1	1	1	1

6.2. A: Power-up Clear

Request:

Start Char	Slave Address	Function Code	CRC
>	00	A	XX

Response:

ACK
A

6.3. C: Set Turn-around Delay

Request:

Start Char	Slave Address	Function Code	data	CRC
>	00	C	0/1/2/3	XX

Response:

ACK
A

6.4. D: Set Watchdog Delay (Digital Board)

Request:

Start Char	Slave Address	Function Code	data	CRC
>	00	D	0/1/2/3/4/5/6/7	XX

Response:

ACK
A

6.5. Set Watchdog Delay (Analog Board)

Request:

Start Char	Slave Address	Function Code	Positions	data	CRC
>	00	D	XXXX	0/1/2/3/4/5/6/7	XX

Response:

ACK
A

6.6. E: Set Protocol

Request:

Start Char	Slave Address	Function Code	data	CRC
>	00	E	0/1	XX

Response:

ACK
A

6.7. G: Configure POSITIONS

Request:

Start Char	Slave Address	Function Code	Positions	CRC
>	00	G	XXXX	XX

Response:

ACK
A



PLEASE NOTE

Positions with a 1 are considered outputs, if no positions are given, FFFF is assumed.
Applies to Analog and Digital Boards.

6.8. H: Configure as INPUTS (unused)

Request:

Start Char	Slave Address	Function Code	Positions	CRC
>	00	H	XXXX	XX

Response:

ACK
A

6.9. I: Configure as OUTPUTS (unused)

Request:

Start Char	Slave Address	Function Code	Positions	CRC
>	00	I	XXXX	XX

Response:

ACK
A

6.10. L: Read Analog Inputs

Request:

Start Char	Slave Address	Function Code	Positions	CRC
>	00	L	XXXX	XX

Response:

ACK	Data	CRC
A	XXXX *#items	XX

6.11. K: Read Analog Outputs

Request:

Start Char	Slave Address	Function Code	Positions	CRC
>	00	K	XXXX	XX

Response:

ACK	Data	CRC
A	XXXX times no of items in Positions	XX

6.12. J: Write Analog Output

Request:

Start Char	Slave Address	Function Code	Positions	Data	CRC
>	00	J	XXXX	XXXX	XX

Response:

ACK
A

6.13. M: Read Digital states

Request:

Start Char	Slave Address	Function Code	CRC
>	00	M	XX

Response:

ACK	Data	CRC
A	XXXX	XX

6.14. K: Write Digital outputs (inputs unaffected)

Request:

Start Char	Slave Address	Function Code	Data	CRC
>	00	J	XXXX	XX

Response:

ACK
A

6.15. K: Write Digital ON (activate)

Request:

Start Char	Slave Address	Function Code	Positions	CRC
>	00	K	XXXX	XX

Response:

ACK
A

6.16. L: Write Digital OFF (de-activate)

Request:

Start Char	Slave Address	Function Code	Positions	CRC
>	00	L	XXXX	XX

Response:

ACK
A

7. General Notes and Troubleshooting

7.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

7.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

7.3. Troubleshooting

7.3.1. Opto22 Internal Error Codes

Our driver uses the following table of error codes internally, they may be issued on the diagnostic screens if an error related to received message content is encountered.

-20	Invalid command number
-21	invalid module
-22	data range error
-23	invalid modifier 1
-24	invalid modifier 2
-25	invalid address
-27	not enough data was returned
-28	invalid data was returned
-31	checksum error on return data
-50	something else wrong with command

7.3.2. Power Supply

The power supply requirements are 5V for digital boards with analog boards requiring a +15v, 0v, -15v supply as well. A standard 5V TTL supply is recommended to ensure correct operation of the brain board.

7.3.3. Enabling Stations

If a remote controller is not enabled, the driver will not communicate with it at all, and the scanned tags for that controller will remain grayed out. Check the setup dialog to see if the controller is enabled.

7.3.4. Enable Tag Saving

The enable tags for the modules are saved in the registry, if the Setup program or the agent server does not have the required security to change the registry, the changes will not be transferred and saved. Change the security on the registry key or log-on as a different user to run Setup or the Agent Server.

Making the registry key

"HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\OPTO22**device**" read-only will stop the saving of the tags, but still allow operator actions on them, these can then be disabled using menu security and putting a password level on the operator actions.

7.3.5. Starting & Stopping Scanning

From adroit version 407 onwards we have implemented a start-stop signal to the driver for polled drivers, this driver will thus not operate correctly with versions prior to build 407.

7.4. Performance

Typical performance indicates over 20 polls per second can be executed, i.e. with 10 modules, the cycle time is approximately 500ms.

If Analog input and output modules are used together, the output modules should be scanned more slowly, since they are polled separately due to protocol differences, this means 2 scans for the one controller, not just one.